Geometric Prequantization

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What is Quantization?

Process of turning a classical theory into a quantum theory.

This means taking observables in a classical theory (whatever that means) to observables in a quantum theory (which are Hermitian operators on some Hilbert space).

Why do we care? Is it even a valid thing to do?

Classical theories tend to be easier to grasp.

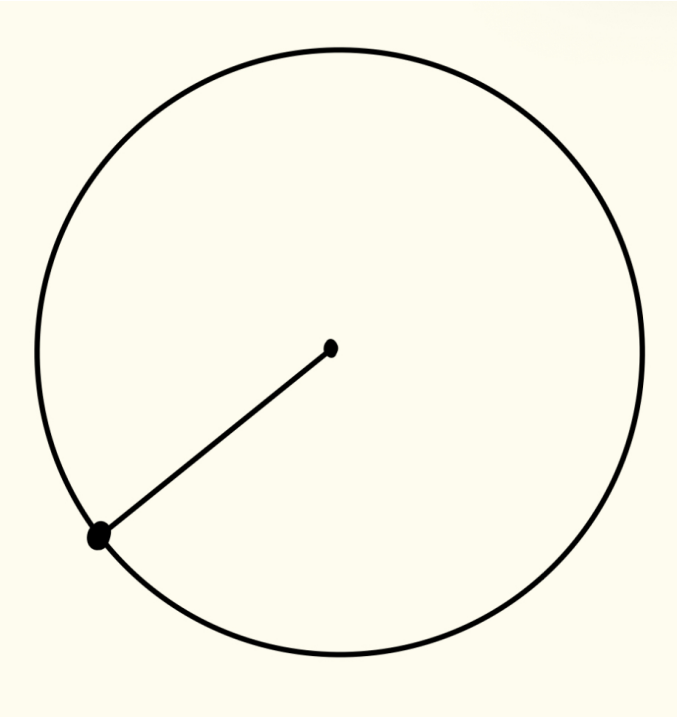
The issue is we are trying to find a general theory through an approximation. It is similar to trying to find a function knowing its value at some approximate limit. Many possible mathematical theories will have the same "classical" limit.

A Brief History of Quantization

- (1900) Planck quantized energy of photons emitted by black bodies to solve the ultra violet catastrophe
- (1905) Einstein quantizes light
- (1925) Heisenberg introduces matrix mechanics
- (1926) Schrödinger introduces Schrödinger's equation through Hamilton Jacobi theory
- (1926) Dirac proposes the "canonical" quantization
- $\vdots$
- (1970) Konstant, Souriau and Kirillov introduce geometric quantization

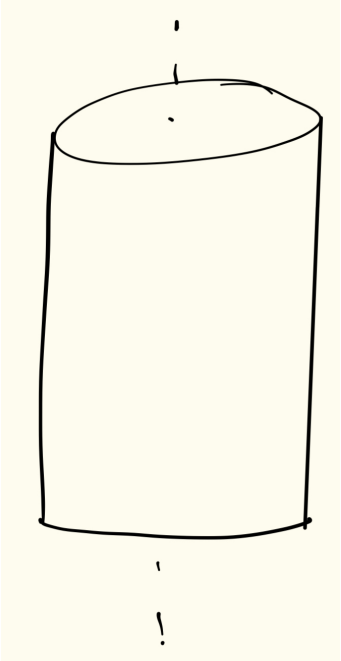
Configuration Space

The configuration space X of a dynamical system is the collection of all possible "position" the system can be in. As an example the configuration space of a pendulum is a circle.



Phase Space

The phase space[1] T^*X of a dynamical system is the collection of all possible "position" and all possible "momenta" the system can be in. As an example the phase space of a pendulum is a cylinder.



[1] For the mathematically inclined, this is the cotangent bundle of the configuration manifold.

Hamilton's Equations

The goal of classical Hamiltonian dynamics is to describe how a system evolves over time in phase space. Mathematically this amounts to finding a suitable system of differential equations satisfied by the positions and momenta.

Similar to how in Newtonian dynamics the positions are governed by a second order differential equation depending on a function F called the "force":

$$\ddot{x} = F(x, \dot{x})$$

Hamilton's formulation is governed by a function called the "Hamiltonian".

Hamilton's Equations

Given a Hamiltonian H the evolution of a position q^i and momenta p_i is given by

$$\dot{q}^i = \frac{\partial H}{\partial p_i}$$
$$\dot{p}_i = -\frac{\partial H}{\partial q^i}$$

Observables

If some system is determined by Hamilton's equations and we care about the measurement of some observable f along the trajectory of the particle then the evolution of f is given by

$$\begin{aligned}\dot{f} &= \frac{\partial f}{\partial q^i}\dot{q}^i + \frac{\partial f}{\partial p_i}\dot{p}_i + \frac{\partial f}{\partial t} \\ &= \frac{\partial f}{\partial q^i}\frac{\partial H}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial H}{\partial q^i} + \frac{\partial f}{\partial t} \\ &= \{H, f\} + \frac{\partial f}{\partial t}\end{aligned}$$

Poisson Structure

The evolution of f now gives us a "bracket" operation on functions given by

$$\begin{aligned}\{f, g\} &= \frac{\partial f}{\partial p_i}\frac{\partial g}{\partial q^i} - \frac{\partial f}{\partial q^i}\frac{\partial g}{\partial p_i} \\ &= \left(\frac{\partial}{\partial p_i} \wedge \frac{\partial}{\partial q^i}\right)(\mathrm{d}f, \mathrm{d}g)\end{aligned}$$

[2]But this is far from the only one!

[2]I like to write it as the second one because it looks scarier than it needs to.

Poisson Bracket

Given a space[3] a Poisson bracket $\{\cdot,\cdot\} : C^\infty(M) \times C^\infty(M) \to C^\infty(M)$ is an assignment so that

- $\{\cdot,\cdot\}$ is linear in both entries.
- It satisfies Leibniz rule $\{f, gh\} = \{f, g\}h + g\{f, h\}$.
- It is antisymmetric $\{f, g\} = -\{g, f\}$.
- It obeys the Jacobi identity $\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0$.

[3] More formally a smooth manifold.

An Example in the Dynamics of a Rigid Body

We can study rigid body motion in the "body frame" where the center of mass is not moving. This makes our configuration space SO(3). We have the commutator on its Lie algebra $\mathfrak{so}(3)$ which then associates to a Poisson bracket on its dual. This bracket is given by

$$\{f, g\}(x) = (\nabla f \times \nabla g) \cdot x$$

Check [7] for more details on Poisson structures on the dual of Lie algebras.

If we look at the Hamiltonian function

$$H(x, y, z) = \frac{1}{2I_1}x^2 + \frac{1}{2I_2}y^2 + \frac{1}{2I_3}z^2$$

we get the equations of motion

$$\dot{x} = \{H, x\} = \frac{I_2 - I_3}{I_2 I_3}yz$$
$$\dot{y} = \{H, y\} = \frac{I_3 - I_1}{I_2 I_3}xz$$
$$\dot{z} = \{H, z\} = \frac{I_1 - I_2}{I_2 I_3}xy$$

An Example in the Dynamics of a Rigid Body

Derived equations of motion for the Euler angles

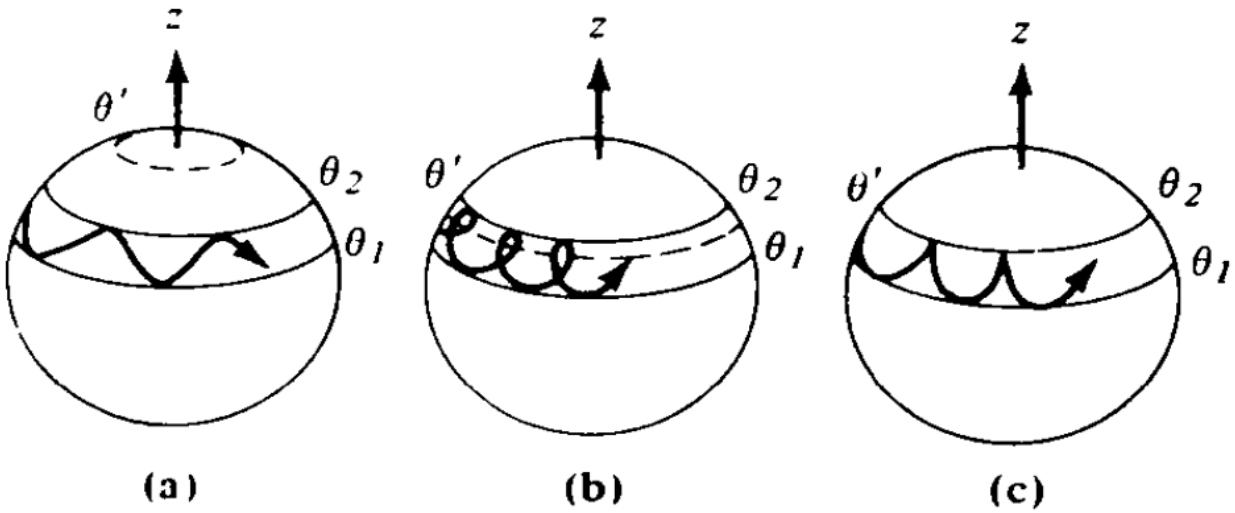


Figure 128 Path of the top's axis on the unit sphere

Check [3].

Symplectic Structure

Along with a Poisson structure, the geometry of phase space is very rich. It comes along with a "canonically defined" differential which in coordinates may be written as

$$\theta = p_i \, \mathrm{d}q^i$$

which gives

$$\omega = \mathrm{d}\theta = \mathrm{d}p_i \wedge \mathrm{d}q^i$$

This is just fancy notation to say "take two vectors X and Y and measure area of the parallelopiped they form in the direction q^i and p_i and add all of them up".

Symplectic Structure

Vector fields on T^*X can now be "inverted" to get differentials using ω. The corresponding differential of X assigns to vectors Y the summed area spanned by X and Y. In coordinates we have that the corresponding differential is

$$X^i \, \mathrm{d}p_i - X_i \, \mathrm{d}q^i$$

where

$$X = X^i \frac{\partial}{\partial q^i} + X_i \frac{\partial}{\partial p_i}$$

Symplectic Structure

Given a Hamiltonian H we can find a vector field X_H whose inversion is the differential of H, we actually want the negative inversion $\iota_{X_H}\omega = -\mathrm{d}H$. This then allows us to talk about the evolution of system as the flow of some vector field. In coordinates we have

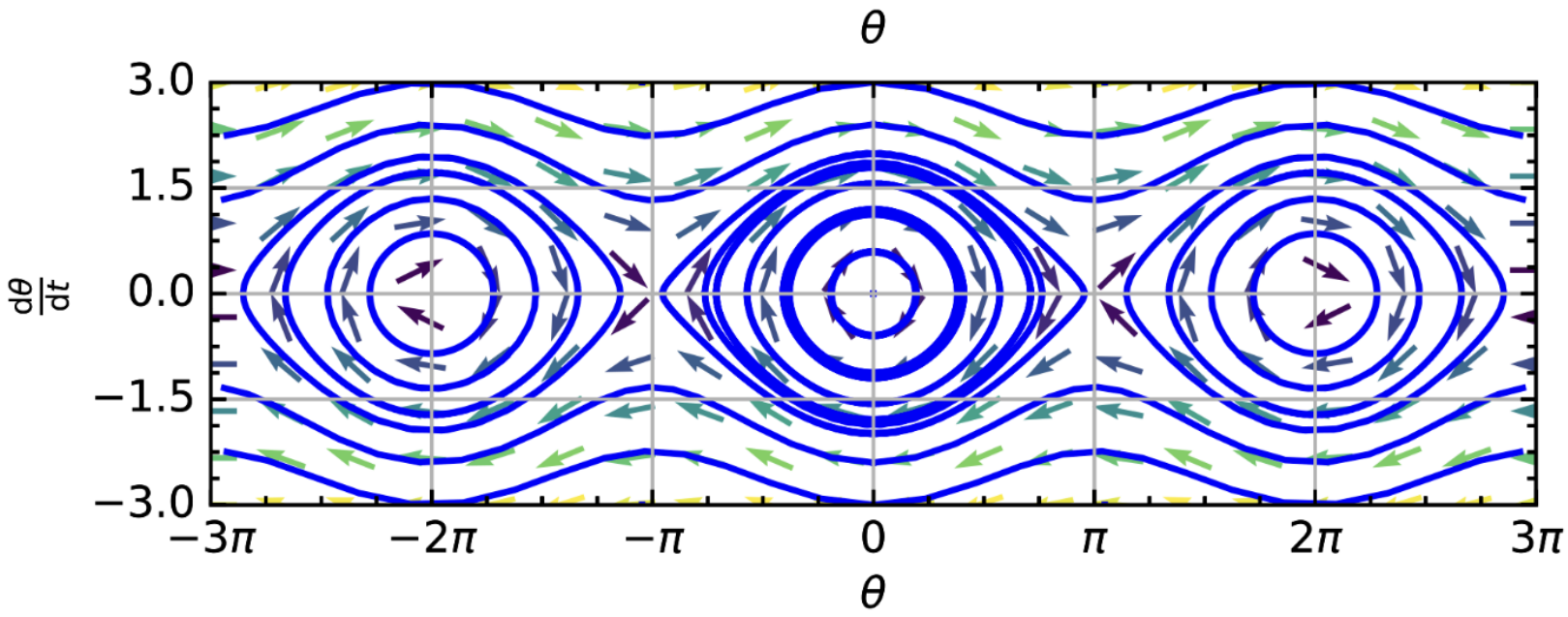
$$X_H = \frac{\partial H}{\partial p_i}\frac{\partial}{\partial q^i} - \frac{\partial H}{\partial q^i}\frac{\partial}{\partial p_i}$$

and so

$$\dot{f} = X_H f = \{H, f\}$$

Hamilton's Motion Via Flow

Dynamics of a pendulum as flow lines in phase space



Check [5]

Canonical Commutation Relations

We now have the following commutation relations between the variables

$$\{p_i, q^i\} = \delta^i_j$$
$$\{p_i, p_j\} = 0$$
$$\{q^i, q^j\} = 0$$

These may look familiar to people who have done quantum mechanics where

$$[X, P] = i\hbar = -i\hbar\{x, p\}$$
$$[X, X] = 0 = -i\hbar\{x, x\}$$
$$[P, P] = 0 = -i\hbar\{p, p\}$$

Noether's Theorem

$$\text{Symmetries} \iff \text{Conserved Quantities.}$$

[4]A symmetry gives a conversed quantity only if $H^1_{\mathrm{dR}}(M) = 0$.

[4]Check out [6] for a detailed proof.

What is Quantization?

We want to send observables in phase space ($f \in C^\infty(T^*X)$) to Hermitian operators[5] on some Hilbert space. We would also like to send Poisson brackets of observables to the corresponding commutator.

[5] One ought to be careful with what Hermicity means in infinite dimensional spaces.

What conditions would we like?

Any "quantization" map Q should (hopefully[6]) satisfy

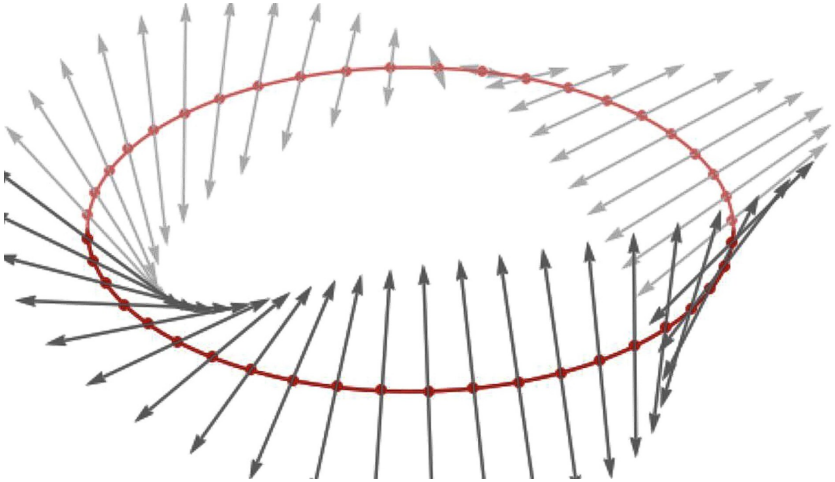
1. Linearity over $\mathbb{R}$
2. $Q(1) = \mathrm{id}$
3. $Q(f)^* = Q(f)$
4. $[Q(f), Q(g)] = -i\hbar Q(\{f, g\})$
5. If $\{f_i\}$ are a collection of observables such that $\{f_i, g\} = 0$ for all observables g implies that $g = \mathrm{const}$. Then $\{Q(f_i)\}$ has the property specifying an eigenvalue for each i we get a unique vector (up to phase) that is an eigenvector for all $\{Q_i\}$ with the specified eigenvalue.

[6] What could go wrong?

Prequantization

Given the phase space of a classical system. What is a canonical choice of Hilbert space?

Mathematicians have already worked out the answer to this! It is the space of "sections"[7] of a particular choice of "line bundle".[8]



[7] Well technically its the completion of this in some larger space.

[8] It comes from wanting the right transformation laws.

Prequantization

After some messing around, one can find that

$$f \mapsto -i\hbar X_f - \theta(X_f) + f$$

satisfies all but the last property. This suggests that the axioms are inconsistent for this very large Hilbert space. See [8] for an example of not being able to quantize properly functions of higher order in p and q.

Prequantization

For generality sake (we will see why we do this with an example) let $\pi : L \to M$ be a line bundle (keep a picture in mind!!). Let s be a trivializing section then a connection may be written as $\nabla s = -i\beta s$ for some $\beta \in \Omega^1(M)$. If $\psi \in \Gamma(E)$ then $\psi = fs$ for some $f : M \to \mathbb{C}$ and $\nabla\psi = (\mathrm{d}f)s - i\beta s = (\mathrm{d}f - i\beta)s$. In this language

$$f \to -i\hbar\nabla_{X_f} + f$$

Math Jargon Prequantization

From a mathematical point of view given a symplectic manifold (M, ω) a prequantization is a choice of Hermitian line bundle $\pi : L \to M$ with connection ∇ such that the curvature R_∇ of the connection is $\frac{1}{\hbar}\omega$.

The existence of such line bundle relies on the Chern class being an integer. This condition is the same as saying that area of any two dimensional closed surface[9] in M is an integer multiple of $2\pi\hbar$.

[9] Up to smooth deformations of boundaries.

Example I of Prequantization

$\mathbb{R}^{2n} = \mathbb{R}^n \times \mathbb{R}^n$ is the phase space of a free particle. We have coordinates $q^1, \ldots, q^n, p_1, \ldots, p_n$ and symplectic form $\omega = \mathrm{d}q^i \wedge \mathrm{d}p_i$. We can take the trivial line bundle $\mathbb{R}^{2n} \times \mathbb{C}$. This gives us

$$q^i \to i\hbar \frac{\partial}{\partial p_i} + q^i$$
$$p_i \to -i\hbar \frac{\partial}{\partial q^i}$$

On wavefunctions that don't depend on the "momenta" this is exactly the Schrödinger representation! We will restrict to this polarization.

Example I of Prequantization

The free particle Hamiltonian is

$$\frac{p^2}{2m} \to \frac{\hbar^2}{2m}\frac{\partial^2}{\partial(q^i)^2} = \frac{\hbar^2}{2m}\Delta$$

and so evolution of wave functions is given by

$$i\hbar\frac{\mathrm{d}\psi}{\mathrm{d}t} = -\frac{\hbar^2}{2m}\Delta\psi$$

Example II of Prequantization

Going back to the pendulum with configuration space S^1 and phase space T^*S^1. In this case we have the area form given by $\omega = \mathrm{d}p \wedge \mathrm{d}\varphi$. The canonical form is given by $\theta = p\,\mathrm{d}\varphi$. Note that the angle function is not a well defined function and so $\mathrm{d}\varphi$ is not the differential of any function.[10]

[10] This can also be seen using Stoke's theorem and the fact that $\int_{S^1} \mathrm{d}\varphi = 2\pi \neq 0$.

Example II of Prequantization

But we can quantize p to get

$$p \to -i\hbar \frac{\partial}{\partial \varphi}$$

which has spectrum $\hbar\mathbb{Z}$.

Example II of Prequantization

Since there are forms that are closed but not the differential of functions, we can therefore deform $\theta \to \theta - \lambda\hbar\,\mathrm{d}\varphi$. This deformation leads to a variety of nonequivalent quantum theories. In this case

$$p \to -i\hbar\frac{\partial}{\partial\varphi} + \hbar\lambda$$

which has spectrum $(\hbar + \lambda)\mathbb{Z}$.

Aharonov-Bohm Effect

The previous example can be seen as encoding the Aharnov-Bohm effect with a magnetic field $\lambda\,\mathrm{d}\varphi$ going through the cylinder. As it predicts that we pick up a phase as we move along the circle.

Whats left?

As we saw in the case of quantization over $\mathbb{R}^{2n}$, we need to restrict our procedure to a certain subspace of the entire Hilbert space. This choice is called a polarization.

Once we have a polarization we can perform a similar produce to produce a theory on this polarization. This later step is called quantization.

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